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Optimization of Analog to Spike Encoding of Sound Data for Sound Source Localization

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Abstract

Sound source localization (SSL) is an everpresent problem in the field of signal processing, a problem whose solutions can be elevated to another level via implementation of neuromorphic principles. In particular, one specific SSL system was already constructed via the use of spiking neural networks, producing results that already rival other neuromorphic methods while requiring zero training [1]. The success and promise of this method only brings even more interest in the possibility of simplifying and enhancing the entire system. This thesis sought to improve the analog-to-spike encoding element via replacing the resource-heavy cochlear model of the ear with a simpler band-gap frequency filter. The alternate method was able to reproduce literature results accurately, albeit at the cost of overall precision. It was concluded that the band-pass filter method showed enough promise in the initial results to warrant further investigation, primarily in the frequency range of the band-pass filter and the time constants of the spiking neurons.

1 Introduction

Sound source localization (SSL) is an incredible feat of the human brain. Although conventional computers can detect sound sources, they require greater resources and higher grade hardware in order to mimic the capabilities of millions of years of evolution. Most practical applications of SSL are in consumer grade products like hearing aids or livestream cameras and therefore benefit from increased efficiency. Thus, it is desirable to keep sound source localization innovations lightweight and simple. Conventional hardware SSL systems rely on large arrays of microphones, high computational power and a non-negligible amount of time to determine sound direction. These three factors make sound source localization for the aforementioned purposes quite inaccessible. Therefore, future applications of SSL would seek novel methods to improve on some of these disadvantages. To rival traditional sound source localization systems, engineers began to take inspiration from the brain. The natural human brain works as a proof of concept to the success of neural networks in power efficiency - they do, after all, calculate and think nonstop in every moment of a person's life.

Neuromorphic computing is a multidisciplinary field which combines aspects of artificial intelligence, neuroscience, physics and electrical engineering to design and manufacture computational systems which operate on a basis more closely related to the human brain than traditional computer systems. One of the innovative methods neuromorphic computing has introduced to the problem of sound source localization is the implementation of spiking neural networks (SNNs) into the process, which allow for quicker and more efficient determination of the direction of arrival (DOA) while keeping resource consumption to a minimum.

This thesis is a follow up to a previous experiment in [1]. In brief, the neuromorphic SSL system in this paper works by converting all analog sound data into a spike train, which is to be fed into the SNN chip. This spiking data is then interpreted by the time difference encoders (TDEs). The TDE is designed to calculate the interaural time difference (ITD) of the microphone array, where the ITD is the difference in time that two ears detect the arrival of a signal. From the ITD, the SSL system is able to determine the DOA. This project focus on an optimization of analog-to-spike encoding method in an attempt to make the entire SSL system simpler and more efficient. More specifically, it will seek to generate spike data for the SNN. The current SSL system uses a NAS analog-to-spike encoding system which takes direct inspiration from the human cochlea. The primary purpose of this paper is to replace this encoder with one specifically targeting the task of SSL, with the hopes of enhancing the system in power consumption and efficiency.

1.1 Research Questions

To summarize, this thesis focuses on answering the following questions:

- Q1: What are some potential analog-to-spike encoding methods that can simplify and make more efficient the current SSL system?
- Q2: How well do these alternative analog-to-spike encoding methods perform in comparison to the original encoding method?

1.2 Thesis Outline

This thesis will focus on investigating the possibility of simplifying and specifying the analog-to-spike encoding methods for the sound source localization system of [1]. This will be done in the following manner:

1. Generate simulated dataset of pure tones
2. Perform the ITD spike encoding via zero counting
3. Record TDE output spikes and compare against different angles
4. Repeat 2. and 3. for the real world dataset
5. Compare simulation results of both datasets to original results

2 Background Literature

2.1 Current state of research

The current field of research in sound source localization focuses on a few varied techniques, each with its own advantages and disadvantages. One traditional example of the more computationally heavy approaches is "beamforming" [2]. In short, beamforming works by reading in all the sound data from each microphone channel, weighting each according to its position in the area, and summing it all together and analyzing this final wave to determine the DOA. It is an effective technique, but overall accuracy and precision relies on a large number of microphones. While it is useful for conventional SSL systems, the two microphone neuromorphic approach of this thesis is not the ideal situation to employ this method. In order to follow the spirit of neuromorphic research, particular focus was given in researching methods of sound source localization using the binaural cues that human brains use, the interaural time difference and the interaural level difference.

In the context of sound source localization in neuromorphic systems, however, the field becomes distinctly narrower. Many of the current neuromorphic methods out there rely on learning and training neural networks, such as [3], which relies on training neural networks to detect sound sources. Some other examples take inspiration from the brain without necessarily resorting to using neural networks, such as [4] which seeks to mimic the human auditory system. While this method does create a more robust sound source localization scheme valid for multiple sound sources of varying quality, it does come with the penalty of being both time and energy consuming to train, which is what this project is trying to avoid. For this reason, the neuromorphic technique that is most appropriate to be employed will involve spiking neural networks. The current state of "neuromorphic auditory models" which use SNNs is summarized in this review paper [5], comparing all the different research groups attempting to create a neuromorphic sound source localization system.

2.2 Spiking neural networks

Spiking neural networks are a type of neural network which take all input data in the form of spikes, in an effort to mimic the way in which the brain works. One of the most remarkable feats of the brain is its low power consumption in comparison to its performance output [7]. Neurons inside the brain communicate between one another via short electric signals, referred to as "spikes". Part of the reason that the power efficiency of the brain is so high comes from the fact that most neurons remain at rest, thereby allowing the spacing and timing of neural activity to encode information in a way that traditional neural networks require much more power to reproduce.

2.2.1 Leaky Integrate and Fire Model

The primary model of the neuron that will be used in this thesis is the leaky integrate and fire (LIF) model. This details of this model can be found discussed in full in [6]. Figure 1, taken from the same source, also illustrates the basic electronic model of the passive membrane.

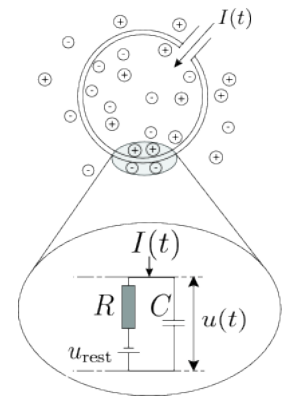


Figure 1: Basic principle of leaky integrator model of the neural passive membrane. Image taken from [6]

The working principle of the leaky integrate and fire model is that the *soma* of the neuron functions in a very similar way to a leaky integrator. The wall of the membrane itself acts as a capacitor for the voltage, and the imperfections of the membrane result in an additional pathway for the current to run. These two elements of the passive membrane means that it can be represented as an RC circuit, and by solving the system of equations implied by this electronic model, one can derive the differential equation that models the dynamic profile of the membrane potential as

$$\tau_m \frac{du}{dt} = -[u(t) - u_{rest}] + R_m I(t), \quad (1)$$

where the time constant $\tau_m = R_m C_m$. For a spiking input of current $I(t) = Q\delta(t - t_0)$, one obtains the following solution to the differential equation:

$$u(t) - u_{rest} = \frac{Q}{C} \exp\left(-\frac{t}{\tau_m}\right) \quad (2)$$

This equation only models the integrating aspect of the model. To simulate synaptic activity, a potential threshold θ was incorporate. This then creates a neuron that, for any membrane potential $u(t) > \theta$, fires a spike down the neural network. This release of a spike then resets $u(t)$.

2.3 Sound source localization

Sound source localization as a topic of interest has been well researched since at least the 19th century [8]. It wasn't until the 20th century when Baron Rayleigh first began working on calculations involving what he called at the time the "binaural ratio". This was the first theory of binaural sound source localization that would later evolve into the modern understanding of ITD and ILD. As noted out in this review [9], our modern understanding of sound source localization follows from both these two binaural cues.

2.3.1 Binaural Cues

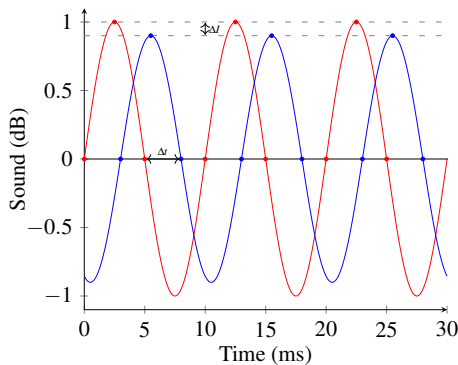


Figure 2: Example of hypothetical sound waves detected by a 2 channel microphone array. Incoming sound wave of first microphone (red) and second microphone (blue). The ITD is represented by Δt and the ILD is represented by ΔI .

The two primary binaural cues in sound source localization are the aforementioned interaural time difference ITD and the interaural level difference ILD. The two cues are used in equal parts by the brain to detect sounds, both effective at low and high frequencies respectively. A visual example of the features in the sound waves that produce these binaural cues is given in Figure 2. ITD-based sound source localization relies on detecting phase differences between two sound waves as they pass through the two ears. This would require that the period of the sound wave remain well above the phase shift to prevent the problem of double counting.

Similarly, ILD-based sound source localization relies on a phenomenon known as "acoustic shadowing" - this is when the wavelength of the sound waves is so short that it must pass through the head of

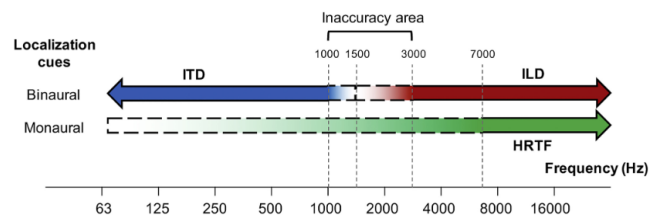


Figure 3: Taken from [9]. Describe the role that both binaural and moonaural cues play in human sound source localization

the human listening instead of around it. The level of acoustic shadowing depends on how long the sound wave needs to pass through the listener's head. This allows the listener to deduce the direction of a sound source based on the intensity difference.

Taking the approximate dimensions of the human head into account, it can be shown that the crossing point for effectiveness in ITD to ILD sound source localization occurs for sound waves of approximately 1500 - 2000 Hz. This balance of SSL cues is illustrated in Figure 2, taken again from the review paper [9].

3 Methods

The process in how this thesis went out to answer the research questions is detailed in this methodology section. The experimental setups and metadata of the datasets used are outlined, as well as the Python packages for the filtering and spike encoding. Finally, this methodology section will discuss a quick outline on how the TDE model works with references to the theoretical basis established in the previous section.

3.1 Experimental setup

In general, for both datasets, the experimental setup remains the same. Figure 4 demonstrates first how the microphone array of both datasets compare to the location of the sound event.

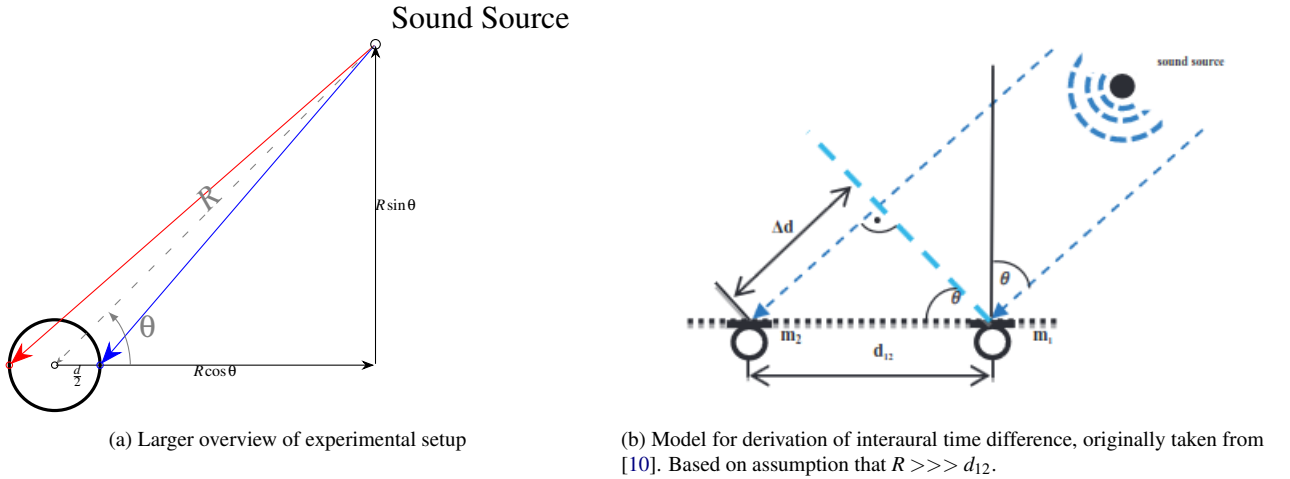


Figure 4: General experimental setup of sound event recordings. **A** : Blue arrow is the sound traveling to the right ear, represented by the blue circle, while the red arrow is the sound traveling to the left ear, represented by the red circle. The angle to be detected is displayed by the gray vector. **B**: Angles shown in this figure refer to the DOA as measured with respect to the front of the microphone array. For this thesis, the DOA will be measured with respect to the *left*.

To simulate a time difference between the two sound waves, a derivation for the interaural time difference was taken from [10]. This model then produces the following formula relating the angle of sound direction (θ) with the detected ITD (Δt). In principle, the sound waves were taken to be far enough away such that they can be modelled as plane waves. The detection of the ITD relies on the knowledge of the interaural distance $d_{12} = 4.8\text{cm}$, as well as the speed of sound $v = 343\text{ms}^{-1}$. The final solution then becomes a simple trigonometric relation involving all three components:

$$\cos \theta = \frac{v}{d_{12}} \Delta t \approx (7146\text{Hz}) \cdot \Delta t \quad (3)$$

3.2 Datasets

This thesis relied on 2 primary datasets, one "perfect" simulated dataset and one noisy synthetic dataset with varying sound type and reverberation, used as a benchmark for the following annual SSL challenge, explained in more detail here[11]. For the rest of this paper, the former dataset will be referred to as the SIM dataset, and the latter dataset as the TAU dataset.

3.2.1 SIM dataset

The SIM dataset amounted to nothing more than 2 sine-waves, representing pure tones with absolutely no reverberation or echo. This ideal dataset was produced as a test to determine the accuracy of the analog-to-spike encoding method as a baseline. In the simulation, it was assumed that the sine-wave originated on the right side of the twin microphone array, and as such the right ear was taken to be the "reference" ear. The two microphones were taken to be 4.8 cm apart, in an attempt to mimic the interaural difference of the microphone setup in the TAU dataset. In order to keep memory issues into consideration when performing the simulation, the smallest non-zero theoretical time difference was calculated. Considering that the time difference decreases as the angle approaches 90° , the time difference expected for a DOA of 80° was calculated using equation 4:

$$\cos \theta = \frac{v\Delta t}{d} \rightarrow \Delta t = \frac{0.048 * \cos(80^\circ)}{343} \approx 0.024ms \quad (4)$$

In order to properly encode this minimal time difference $\Delta t = 0.024ms$, a sampling rate of at least $\frac{1}{0.024}ms^{-1} = 41667Hz$ was needed; 42000Hz was chosen. With these two varying parameters, this resulted in 400 different samples being created, each with a "reference" wave and a "secondary" wave, where the secondary wave contained the phase shift and lower amplitude.

The simulation was ran on the SIM dataset in order to provide a baseline accuracy for future real world experiments. Firstly, both binaural cues were tested for frequencies in the range from 500 Hz to 5500 Hz in 500 Hz increments, resulting in the testing of different pure tones of multiple frequencies. Additionally, the DOA was tested from the range of 0° to 80° in 5° intervals.

3.2.2 TAU dataset

Microphone positions	$[45^\circ, 35^\circ, 4.2cm], [-135^\circ, 35^\circ, 4.2cm]$
Sound Type	Varied from 13 different classes
Elevation	$[-40, 40]^\circ$ in steps of 10°
Azimuth	$[-180, 180]^\circ$ in steps of 10°
Radius	1 m, 2 m

Table 1: General metadata of TAU dataset. The positions of the microphones are given in spherical coordinates (ϕ, θ, r) , where ϕ represents the azimuth (polar angle), θ represents the elevation and r represents the radius as measured from the center of the EigenMike.

The TAU dataset was originally taken from an online repository at [11]. Table 1 describes the general features of the dataset. The original use of the dataset was primarily reported on in [12] and [13], with a technical report of the data available in [14]. In summary, the data involves "multichannel recordings of sound scenes in various rooms and environments" [11]. The data involved 4 splits of 100 sound files, each file containing approximately 20-50 sound "events". In the interest of keeping the calculation time manageable to the deadlines of the project, the entire dataset was not used; only the *first* training split was used. This amounted to 100 sound files, which corresponded to 3876 total sound events. The accompanying ground truth of the dataset contained the categorization of each sound event into classes, the start and end times of the event in particular and the location of the sound source in spherical coordinates. The original dataset involved recording on an Eigenmike, a 32-channel spherical microphone. Only two channels were used, whose positions can be seen in

Table 1.

A band gap filter was applied on all the incoming soundwaves before any spike encoding was performed. This allowed for the less relevant frequencies of the sound sample to be filtered out and allowed for the target frequencies to be converted into spikes. An order-ten Butterworth band-pass filter of 480Hz to 520 Hz was chosen for this task. Figure 5 shows a simple gain response of generic butterworth filters of varying orders. In order to remove the phase delay inherent in signal filtering, the signal was passed forwards and backwards. After filtering, they were converted into spikes and then sent to the TDE.

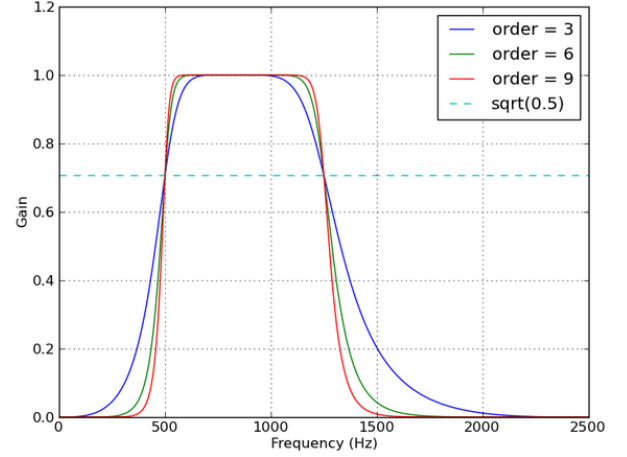


Figure 5: Example of a Butterworth band-pass filter, originally taken from Stack Overflow [15]. Ideally, all frequencies outside the bandpass pass with a gain of 0, while frequencies within pass with a gain of 1. The increasing order of the Butterworth filter dictates how strictly this ideal case is followed.

3.3 Python Packages

The main programming language used in creating the model was Python. For processing the raw data, the primary packages used were NumPy [16] and SciPy [17]. In particular, the PyTorch [18] library was used extensively in the simulation of the TDE. SciPy was mainly used for its audio-processing capabilities, in particular reading .WAV files and applying frequency filters. NumPy was useful in mainly for the use of NumPy arrays, which made the handling of the large amount of data used in this thesis manageable and intuitive. Finally, the graphs in the results were constructed using Matplotlib.

3.4 Analog-to-spike encoding algorithms

The following section below will describe the entire analog-to-spike conversion process, beginning with the original spike encoding from the sound data itself. Firstly, the specific algorithm chosen to translate the information concerning the ITD of the two sound waves into spikes is discussed. Then the Python model for the TDE is explained in more detail.

3.4.1 Interaural Time Difference

Interaural time difference encoding was accomplished via counting zero crossings. The advantage of counting each zero crossing is that their timestamps are exclusively functions of frequency. Since the frequency profile of the soundwave is invariant across the two microphones, the relative distance of the two zeros become measurements of the ITD. The actual counting of the timestamps of the zero crossing was done in Python through counting sign changes. For every sign change counted, the timestamp of said "zero-crossing" t_z was recorded (in seconds). This was done for both incoming sound waves. The spike train was then produced using these timestamps, by following the simple conditional formula below,

$$\text{Spike at index } a = \begin{cases} 1, & \text{if } a = t_z \cdot \text{SR} \\ 0, & \text{otherwise,} \end{cases} \quad (5)$$

where SR represents the chosen sample rate (42kHz for SIM, 48kHz for TAU). This spike encoding provides the spiking input to then be fed into the TDE, which then theoretically should produce spike proportional to the ITD - the specifics of which are detailed in the following section.

3.4.2 Time difference encoder model

In order to make sense of the encoded zero-crossings, a TDE was used. This was modelled after the one used in [1], and therefore would be expected to produce similar results. Figure 6, taken from [1], gives a quick understanding on it's working principles.

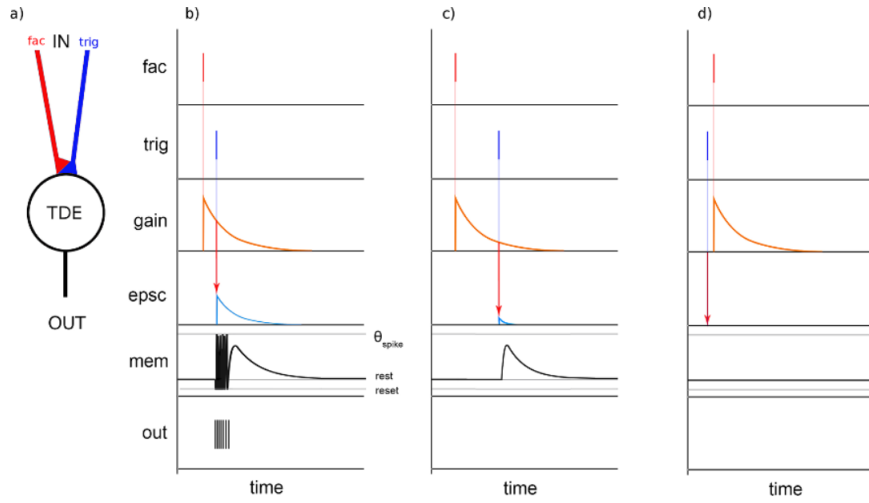


Figure 6: Working principle of the TDE, taken originally from [1]. The facilitatory ("fac") and trigger ("trig") inputs the spike inputs derived from the first and second microphone channels. The "fac" inputs prepare the exponentially decaying gain, which is then converted into excitatory post-synaptic current (EPSC) by the "trig" input. EPSC is passed into the LIF neuron where it is integrated over in the "mem" plot, which then produces spike bursts as it's output ("out"). The three different columns represent 3 different potential arrival times of the "fac" and "trig" spikes. **A**: "trig" arrives soon after "fac", therefore small Δt . **B**: "trig" arrives long after "fac", therefore long Δt . **C**: "trig" arrives before "fac", therefore outside angle range of TDE

The range of DOA azimuths chosen to be analyzed were only in the range of 0 to 90 degrees with respect to the "left hand side" of the microphone array i.e. the values to the left of the first microphone located at $(45^\circ, 35^\circ, 0.048m)$. The reason that this limitation was put in place lies in the known limitations of binaural sound source localization. Since only 2 microphones were used, and the ITD is a scalar value depending on the distance, it is invariant under the Cartesian transformations $x \rightarrow -x$ and $y \rightarrow -y$. Therefore, all significant data can be derived from simply analyzing one quadrant of angles, most conveniently $[0, 90]^\circ$.

The model simulates the TDE via Euler method of solving the differential equation 1 for each timestep. The time constants of the facilitatory, trigger and membrane potentials, respectively τ_f , τ_t , τ_m , were free to be set differently from one another but were all chosen as the same value.

4 Results

The results of the simulation as described above are outlined below, organized per dataset.

4.1 SIM dataset

The following results represent the simulations for the SIM dataset. These results serve as a benchmark for comparison to real world data. Seeing the spiking behaviour of the TDE on "perfect" synthetic pure tones as well as the limitations of the encoders establishes a maximum precision benchmark to compare with noisier data. Detailed below are the techniques used to determine the average spikes per burst, as well as the results of the TDE's output spikes compared to both azimuth angle and ITD.

Figure 7 describes the modeled relationship between the interaural time difference and the incoming DOA. The black points on the red line show the values taken off that were then encoded into the simulated sound waves. The reason this relationship is important to know is that it allows for a formula between the angle and the ITD to be established. In the TAU results, there will only be comparison to the angles, since they are already provided in the metadata of the dataset. For this reason, in order to be able to compare the literature results, which compare number of spikes N_s with time difference Δt , it is necessary to produce a relationship between Δt and the incoming angle θ .

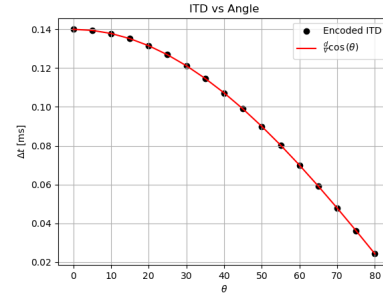


Figure 7: ITD plotted against DOA. The red line represents the theoretical prediction based on the equation of the model in Figure 4.

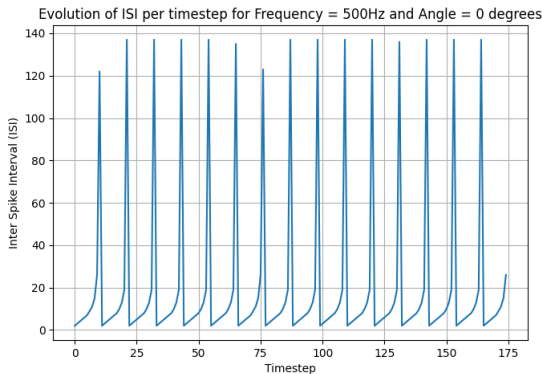


Figure 8: Interaural spike intervals against time. Note that timesteps represent time of the ISI distances, not of sound waves

erage spikes per burst it is sufficient to show that

In order to get the average spiking rate of each *burst* of spikes, a method in determining number of bursts was required. The way this was done in the thesis was via measuring the inter spike interval (ISI) of each of the spikes, illustrated in Figure 8. The ISI is inversely proportional to the spike rate of the LIF neuron, and the spike rate is directly proportional to the EPSC produced by the TDE. Since the EPSC produced by the TDE is monotonically decreasing by construction, the monotonically increasing ISI would only drop in the presence of a new spike burst. This means that each local maximum of the ISI plot reveals the presence of a burst, and to determine the av-

$$\tilde{S} = \frac{N_s}{N_m}, \quad (6)$$

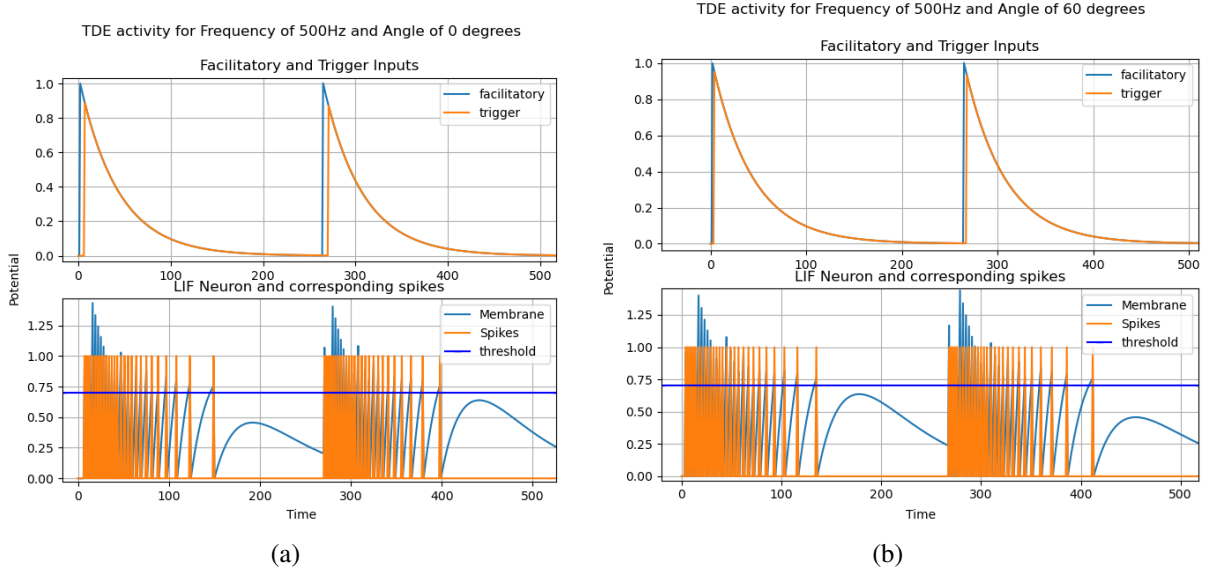


Figure 9: Top half: Plots of Facilitatory and Trigger output simulations. Bottom half: Plots for LIF membrane potential and spike outputs, with potential threshold of LIF neuron highlighted. All time constants held to the same value of $\tau_f = \tau_t = \tau_m = 0.41ms$.

where $\bar{S}$ represents the average spike count, and N_s, N_m represent the number of total spikes and total ISI maxima respectively.

As the TDE simulations were run, both the number of bursts and the total spikes were counted. The following plots show some examples of the simulation of the TDE. The increase of the angle from 0° to 60° resulted in the decrease of the interaural time difference, as seen in the distance between the facilitatory and trigger plots. Although the TDE was simulated for multiple frequencies between 500 Hz and 5500 Hz, it only produced significant results for the lower frequencies. Figure 9 displays the results for 500 Hz. Interestingly enough, this mimics the same limitation human beings face in ITD sound source localization, where higher frequency sounds render the ITD harder and harder to resolve. An example of this TDE behaviour is left in the appendix for the interested reader.

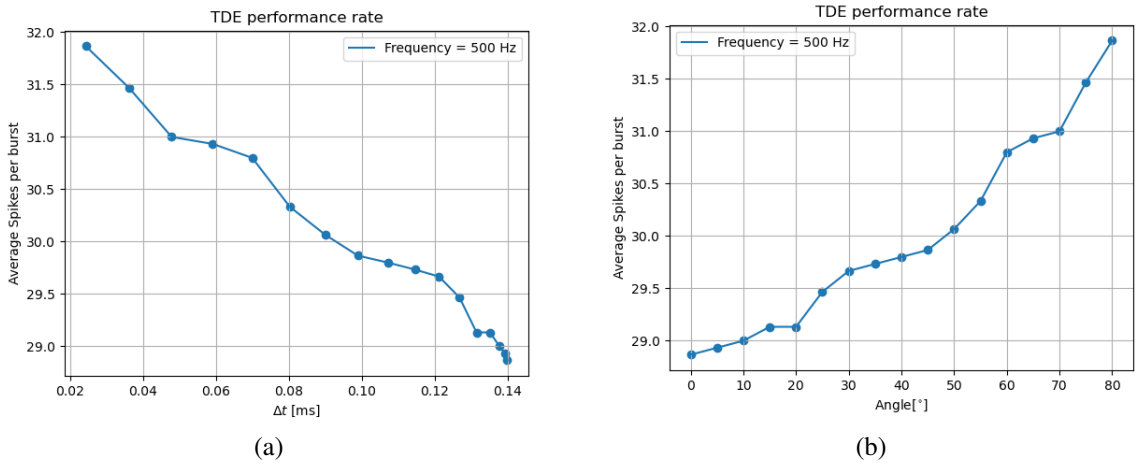


Figure 10: Evaluation of average output spikes per spike burst as a function of **A)** ITD and **B)** DOA. For similar plots for higher frequencies, see appendix

Finally, Figure 10 show the final desired results of the simulation. The average number of spikes per burst were inversely proportional to the recorded ITD, while being proportional to the incoming angle. This was the expected relation, as seen before in Figure 3.

4.2 TAU dataset

The inconvenient reality of analog audio recordings is the general limit of the amount of information that can be encoded. The sample based microphone used in the TAU dataset take timesteps at a rate of 48kHz. While this is sufficient for most applications, such a sample rate becomes problematic in the instance when one is counting zero crossings. Since the interaural distance of the microphones is relatively small, the corresponding time difference would result in ITDs too short to be properly read by the microphone. Figure 11 shows an example of the inverse of the predicted time difference (Δt^{-1}) compared to the DOA for the TAU dataset. The horizontal blue line represents the native sample rate of the microphones, and therefore also serves as a threshold for resolvable ITDs, where all angles with ITD values above this threshold are not resolvable due to limitations of recording hardware.

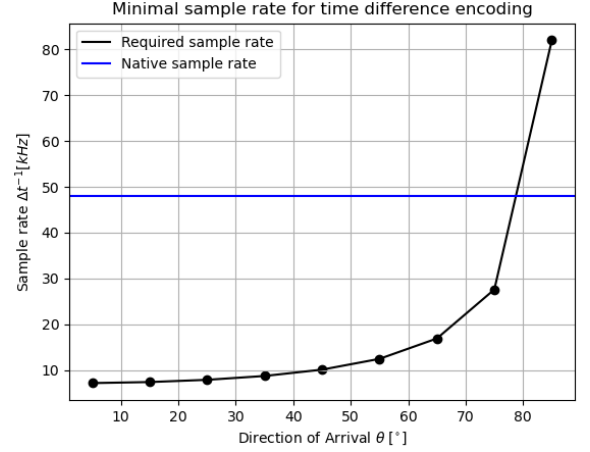


Figure 11: Direction of arrival and inverse of corresponding time difference.

As described before in the methodology, the sound wave required filtering before spike encoding could occur. An example of the filtered sound wave is given in Figure 12, with the zero crossings in the filtered wave marked down for clarity.

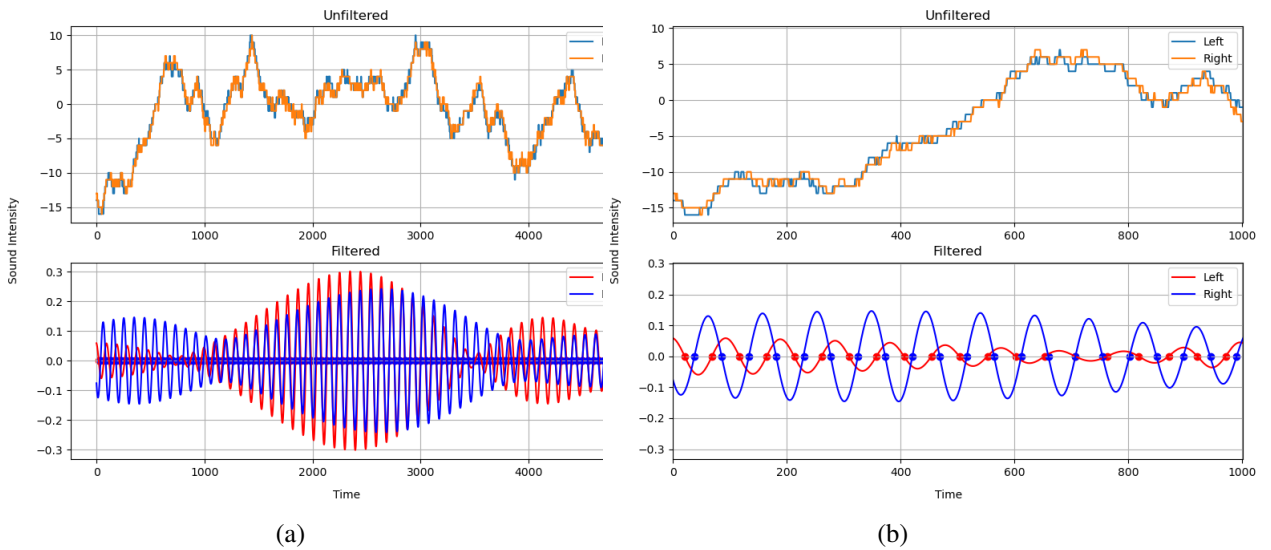


Figure 12: Filtered vs Unfiltered sound data for time range of **A**: 0 to 100 ms and **B**: 0 to 20 ms.

After the sound file was successfully filtered, the zero crossings of both channels were counted and converted into spike trains on PyTorch tensors. These spikes were passed into the model of the TDE

and the resulting spike activity parameter A_s was recorded alongside the ground truth metadata of that particular sound event - that is to say, information of the sound class, the elevation, the distance and most importantly the azimuth.

4.2.1 TDE output

Since the entire experimental setup has an angular offset of $\theta_{off} = 45^\circ$, a rotation of the reference frame was required. The formula to transform the recorded metadata angles θ_{meta} to the DOA angles we desired θ_{doa} was thus:

$$\begin{aligned}\theta_{doa} &= (90^\circ - \theta_{meta}) - \theta_{off} = 45^\circ - \theta_m \\ \theta_{doa} \in [0, 80]^\circ &\rightarrow \theta_{meta} \in [-35, 45]^\circ\end{aligned}\quad (7)$$

With this angular range, the final slopes were calculated for the TAU dataset. The errors are calculated from the variance of the average spikes per burst $\tilde{S}$. The line of best fit was calculated using the method of least squares for a linear fit. The time constant used was kept the same as in the SIM dataset, $\tau_f = \tau_t = \tau_m = 0.41ms$.

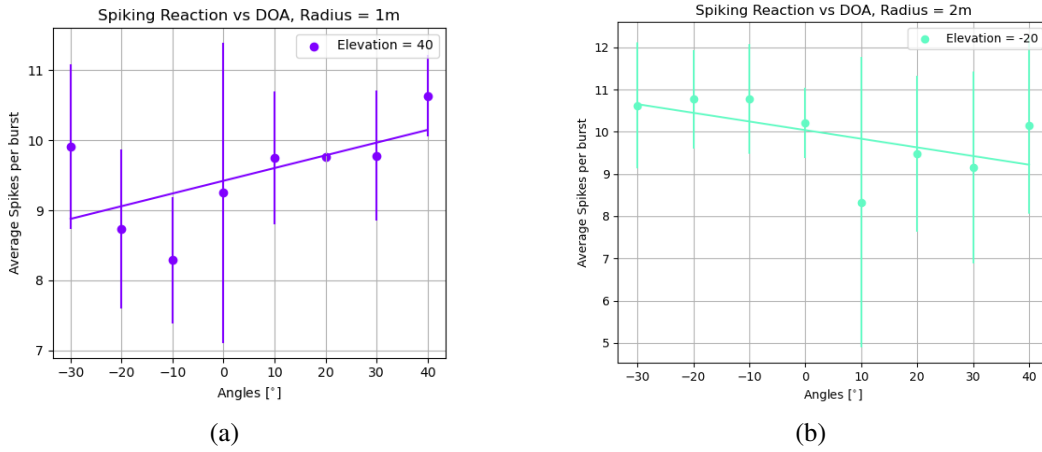


Figure 13: Output spikes for TAU dataset. Errors shown represent 1σ

In Figure 13, the results of the highest elevation of the one meter distance dataset were shown next to the results of the lowest elevation for the two meter distance dataset. This was done to demonstrate the general diversity of results in the final simulation. For the first graph, one notices quite a similar trend to the SIM dataset, which would suggest some success in the filtering process of replicating the OpenNAS results. However, for the second plot with -20° elevation, the inverse trend appears. It is useful to note that variation in elevation is a factor of the TAU dataset that is reproduced neither in [1] nor in the SIM dataset. The implications for this variation in the output would suggest that the elevation of the incoming sound source is not negligible in the resulting ITD as previously thought.

5 Discussion

5.1 Comparison of elevation and distance to expected results

When producing the final plots of the dependency of the spike rate on the *azimuth* of the DOA, special care was taken to organize the datapoints by elevation and distance. The interesting part of all of this is that there appeared to be some kind of dependence on the spike to angle slope with respect to the elevation.

Since the two microphone array lies on the exact same elevation, one would expect there to be a symmetry along this plane - in otherwords, an invariance on the Cartesian transformation $z \rightarrow -z$. Therefore, positive and negative elevations should theoretically result in similar spiking reactions from the TDE, resulting in an expected quadratic relation. Figure 14 shows the final expected values:

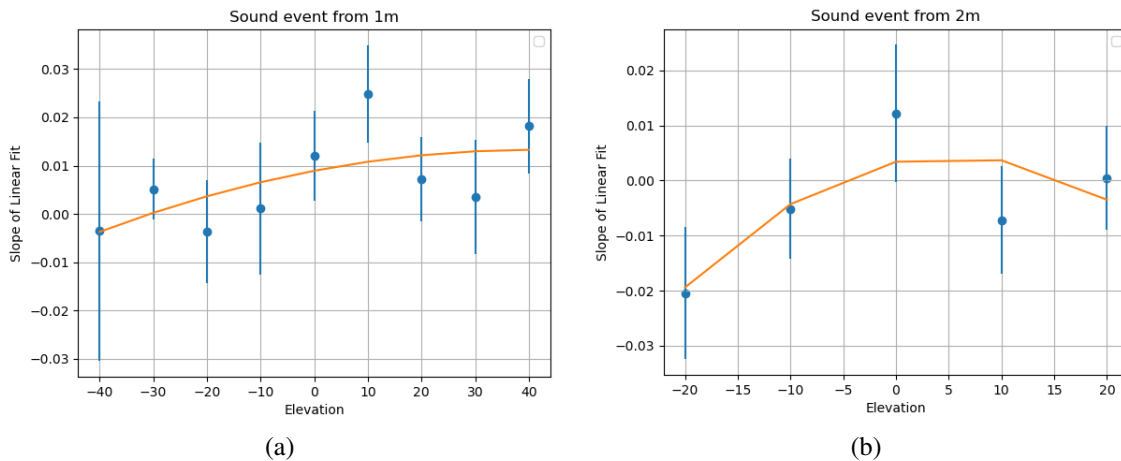


Figure 14

Although the line of best fit plotted was quadratic, it can be qualitatively concluded that the trend resembles closer a linear fit. To rephrase, this means that the expected planar symmetry is not appearing in the final results. This alone warrants further investigation into the effects of elevation on azimuthal DOA, but there exists two possible explanations for this final result that don't assume outside of the theoretical predictions.

One possible explanation for this is due to the nature of the recording device itself. The original recording device, the EigenMike, consisted of 32 sound channels placed evenly all around the spherical surface. It was never intended to have 2 specific channels taken out and used independently. For this reason, it could be that the different results in the elevation are due to an inaccuracy of capturing the recordings below the microphone array.

Secondly, this result could also be a problem that resolves itself depending on the distance of the sound event. Comparing the first curve to the second, one sees that the expected quadratic shape begins to emerge in the second graph. Further samples from higher distances are needed in order to draw any concrete conclusions.

5.2 Similarities and differences of reproduced literature results

One of the biggest differences between the obtained results of this simulation and the reference results by Schoepe et al. is the diminished spiking range of the TDE. This was noted in both the SIM and the TAU datasets. While this is not a fault of the TDE model, the discrepancy is worth investigation. The reason for this is concluded to be most likely a problem with the sample rate and the experimental setup of the data used.

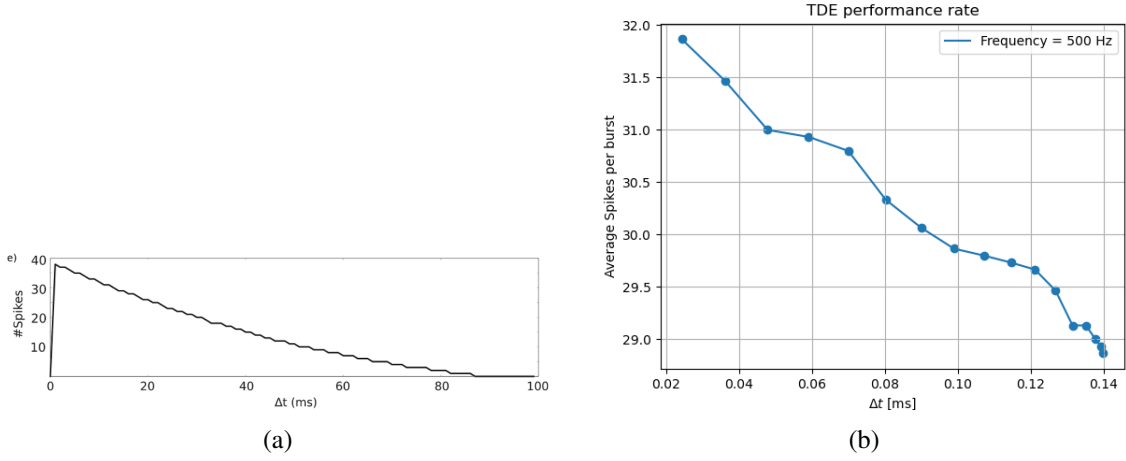


Figure 15: Comparison of **A)** real world results with **B)** SIM dataset results. Success of SIM model give further evidence to equation 3.

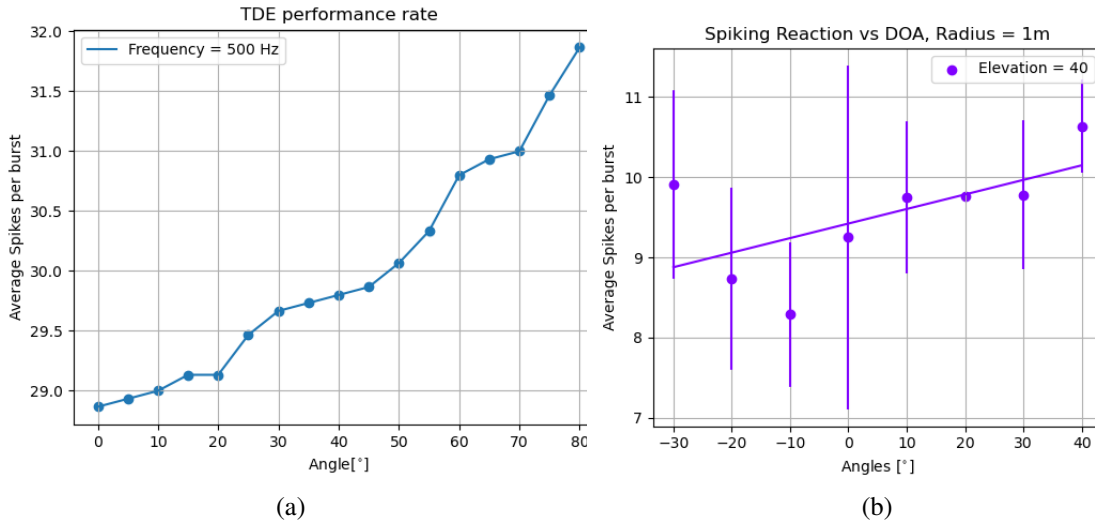


Figure 16: Comparison of spikes per burst against angle for the **A:** SIM dataset and **B:** TAU dataset.

One of the bigger drawbacks of the real world dataset is the distance between each microphone. The interaural time difference Δt is directly proportional to interaural distance d . Theoretically, the exact magnitude of the ITD only matters in relation to the incoming sound frequency f , such that $\Delta t \ll \frac{1}{f}$ must be satisfied. However, in practical applications, the sampling rate of the ITD must be considered as well. If the difference between the ITD of two different angle samples is smaller than the timestep of a specific sample rate, then it is impossible for the system to resolve the ordering of the two spikes since all periodicity of the waves is lost. This was seen in Figure 11, where even before any zero

counting was done the TDE was already barred from some angle ranges. Even if this requirement is fulfilled, if the ITD is too small in relation to the timestep of the sample rate, then the sensitivity of the entire simulation is affected. The Euler method used to simulate the LIF neuron can only produce the desired spikes at the corresponding timesteps, so there exists a limiting factor in there not being precise enough timesteps to properly simulate a continuous system.

The simplest solution is to simply use an experimental setup where the interaural distance are greater - this also makes sense from a neuromorphic perspective, as the size of the human head is larger than the 4.8 cm used in this project. It is possible to simply design higher and higher sample rate microphones, but this will only lead to complicating a system that this thesis seeks to simplify.

6 Conclusion

The original goal of this thesis was to investigate the possibility of replacing the cochlear model of the ear with a simpler frequency filter. This enhancement was accomplished, albeit with mixed results. Although the general trend and accuracy of the results were maintained, the overall precision suffered as a result of the less stringent filtering system. So, it is concluded that it is indeed possible to replace the OpenNAS with a simpler filtering method. However, in practical terms, to fully replace the entire system would require more investigation into the exact filtering method to be employed in such a way that the precision of the system can be improved.

Some specific areas in which the filtering system can be improved involve changing the band-pass of the filter, both by adjusting the magnitude of the frequency as well as the width of the band-pass. Additionally, the simulations were all carried out using only one time constant $\tau = 0.41\text{ms}$. A future study looking to build upon this project could begin by studying the effect of these frequency bands and time constants on the precision of the results.

6.1 Summary of Main Contributions

This work presents the first real look in simplifying and starting to expand upon the research presented in [1]. The main contributions can be summarized into the following points:

1. The first methodological attempt at replicating and replacing the OpenNAS of the original system with a simpler band-pass filter was carried out.
2. Testing the SSL system against larger variety of sound events, including variation in elevation and distance.

6.2 Future Work

One of the shortcomings of this project is that the real world data included information on varying distance and elevation, but the sound source localization of this project was only built to determine the azimuth of the sound source direction. For real world applications, an expansion on this thesis could be to research specific elevation and distance encoding methods.

There already exist studies which investigate this, primarily by explaining the biological solution that blind human beings use to determine vertical location [19]. To summarize the findings, vertical sound source localization in humans come about as a result of the *distortion* of the incoming sound, created via diffraction of the sound wave inside the ear. A potential future extension of this project could be to investigate replicating and encoding this distortion into spikes. Using this hypothetical distortion encoder alongside the current TDE would bring the SSL system from two dimensions to three, giving the entire system more practicality in the real world.

Appendices

A Additional plots

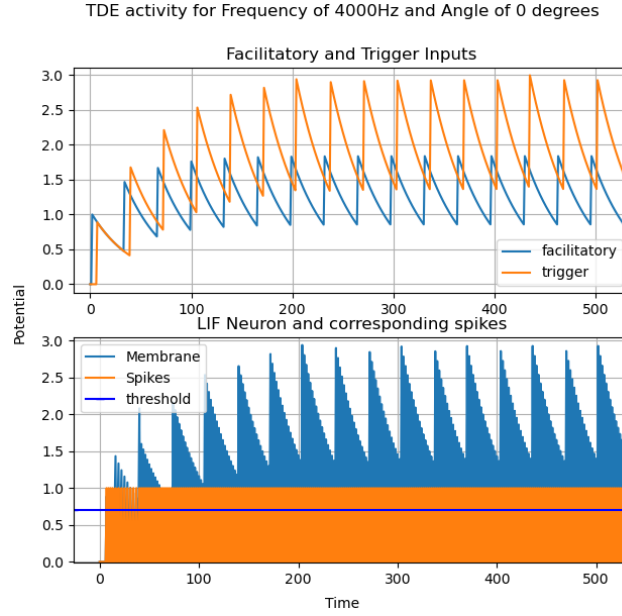


Figure 17: Similar plot as above, but frequency of incoming pure tone is 4000Hz. The increased frequency means increased rate of facilitatory spike inputs, which create an increased gain for the trigger’s EPSC current and lead to uncontrollable spiking.

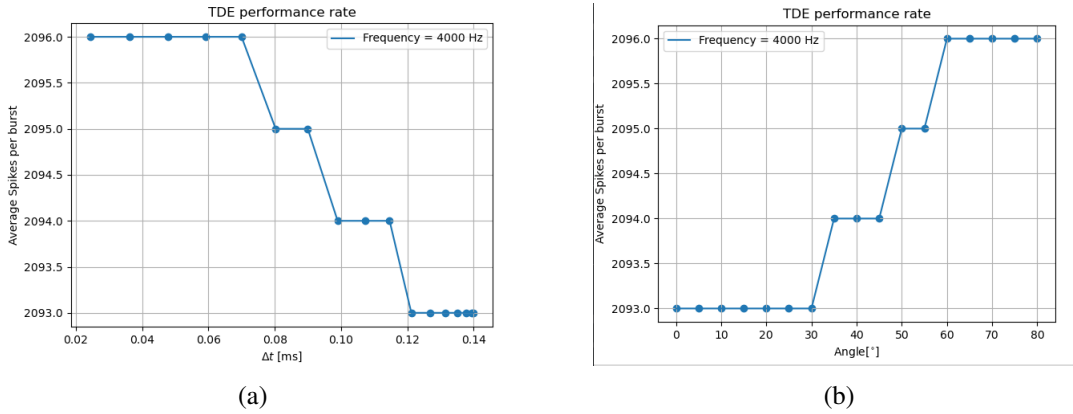


Figure 18: Final spike output plots for 4000Hz pure tone. Only 4 distinct ”spikes per burst” values means that this frequency range is not suitable for one-to-one spike encoding of the ITD

B Code and dataset

The python code used in this thesis is freely available in GitHub [here](https://github.com/ioannistassioulas/bachelorproject).

(URL: <https://github.com/ioannistassioulas/bachelorproject>). Said link also contains the link to reproduce the SIM dataset. The TAU dataset can be found in the sources as cited in the thesis, namely [12], [13], [14] and the [website](https://dcase.community/challenge2022/task-sound-event-localization-and-detection-evaluated-in-real-spatial-sound-scenes) [11]. (URL: <https://dcase.community/challenge2022/task-sound-event-localization-and-detection-evaluated-in-real-spatial-sound-scenes>)

Glossary

DOA Direction of Arrival. 5, 7, 10, 11, 14–16, 18

EPSC Excitatory Post Synaptic Current. 13, 14

ILD Interaural Level Difference. 8, 9

ISI Inter Spike Interval. 14

ITD Interaural Time Difference. 5, 6, 8–10, 12–17, 19, 20, 22

LIF Leaky Integrate and Fire. 7, 13, 14, 20

SNN Spiking Neural Network. 5, 7

SSL Sound Source Localization. 5, 7, 9, 21

TDE Time Difference Encoder. 5, 6, 10, 12–14, 16, 18–21

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